

The scalable section-domain route

Removing the 2^B axis, guaranteeing energy, and learning why local evidence did not decode complete messages

Stage	4 of 5
Updated	5 September 2026
Status	Implementation and bridge tests complete; current outer-code/decoder route did not retain $L = 1$ performance

1 Why this route was attempted

Report 3 avoided storing $\Phi \in \mathbb{C}^{n \times 2^B}$ but retained the length- M maps T_ℓ , the count vector a , and the adjoint score $\Phi^H r$. At $B = 100$, this is still impossible. The section-domain idea was to generate a legal tuple of small local indices directly from the payload and to decode only local occupancies.

This is closely related to the outer-code stage of coded compressed sensing (CCS): payload information is distributed over section symbols and parity constraints rule out most arbitrary combinations. The framework deliberately separated this outer validity map from the physical local banks so alternatives could be tested. Nevertheless, any concrete section code is a model-class restriction. The experiment therefore needed bridges back to the unrestricted $L = 1$ landscape, not only a proof that large- B tensors fit in memory.

2 From the explicit equation to a message-axis-free encoder

The explicit model is

$$y = \sum_{\ell=1}^L F_\ell T_\ell a + z, \quad F_\ell = B_\ell U_\ell \in \mathbb{C}^{n \times N_\ell}.$$

Define local occupancies

$$s_\ell = T_\ell a \in \mathbb{Z}_+^{N_\ell}.$$

If the transmitter can compute each selected index $i_\ell(w)$ procedurally from message bits $w \in \{0, 1\}^B$, it does not need to store the $N_\ell \times 2^B$ matrix T_ℓ . For active messages $w_1, \dots, w_{K_a}$,

$$s_{\ell,j} = \sum_{u=1}^{K_a} \mathbf{1}\{i_\ell(w_u) = j\}, \quad y = \sum_{\ell=1}^L F_\ell s_\ell + z.$$

The state dimension becomes $\sum_\ell N_\ell$. For $B = 128$, $J = 16$, and $L = 16$, the current design uses $N_\ell = 2^{16} = 65,536$, hence 1,048,576 local coordinates rather than 2^{128} global coordinates.

The selected waveform for one message is

$$\phi(w) = \sum_{\ell=1}^L F_\ell[:, i_\ell(w)].$$

This is a genuine scalable forward representation if the path and selected columns are generated without enumerating all messages. It does not yet imply that the receiver can infer w from the aggregate local evidence.

3 Exact energy after training and deployment

URA normally fixes the energy of every codeword, not merely the average energy of sampled training messages. If arbitrary full-frame section atoms overlap, then

$$\|\phi(w)\|_2^2 = \sum_{\ell} \|f_{\ell, i_\ell}\|_2^2 + 2\Re \sum_{\ell < k} f_{\ell, i_\ell}^H f_{k, i_k},$$

so normalising each local column alone does not control the cross terms for unseen paths.

The implementation therefore provides an exact orthogonal-subspace construction. Put section bank $G_\ell \in \mathbb{C}^{n_\ell \times N_\ell}$ into its own latent coordinate block, choose energies $E_\ell \geq 0$ with $\sum_\ell E_\ell = 1$, and optionally apply a fixed unitary mixer Q_{mix} :

$$\phi(w) = Q_{\text{mix}} \begin{bmatrix} \sqrt{E_1} G_1[:, i_1(w)] \\ \vdots \\ \sqrt{E_L} G_L[:, i_L(w)] \end{bmatrix}, \quad \|G_\ell[:, j]\|_2 = 1.$$

Because $Q_{\text{mix}}^H Q_{\text{mix}} = I$,

$$\|\phi(w)\|_2^2 = \sum_\ell E_\ell \|G_\ell[:, i_\ell(w)]\|_2^2 = 1$$

for every possible message, including combinations never sampled during training. After each optimiser step, projecting every local column back to unit norm preserves the deployment guarantee.

The mixer can make physical coordinates look dense while preserving norms, inner products, and white-noise geometry. It is not innocuous under resource-dependent fading or a requirement for literal zeros. The alternative overlapping full-frame mode uses a sampled path-energy penalty; that can improve training but cannot certify unseen messages. The exact mode was therefore the primary route.

4 Procedural outer code and parity matrix

4.1 Information symbols and redundancy

Group B payload bits into $k = B/J$ symbols

$$u = (u_1, \dots, u_k) \in \mathbb{Z}_q^k, \text{quad}q = 2^J.$$

The current arithmetic is modulo q . It is correctly denoted $\mathbb{Z}_q$; for $J > 1$, $\mathbb{Z}_{2^J}$ is not a field and should not be called $\text{GF}(2^J)$. The encoder adds r parity symbols and returns

$$x = (x_1, \dots, x_L) \in \mathbb{Z}_q^L, \text{quad}L = k + r.$$

Parity adds transmitted sections and channel resources, but not payload bits. With $B = 128$, $J = 16$, $k = 8$, $r = 8$, the same 2^{128} payloads are represented by 16 section symbols. The unconstrained tuple space contains $q^L = 2^{JL}$ paths; the parity equations select $q^k = 2^B$ valid ones. Therefore $L2^J$ is the number of local states, not the number of complete messages.

4.2 How the parity matrix stores checks

Each check c is a modular equation

$$\sum_{\ell \in \mathcal{N}(c)} H_{c\ell} x_\ell = 0 \pmod{q}.$$

Stacking r equations gives

$$\boxed{Hx = 0 \pmod{q}}, \quad H \in \mathbb{Z}_q^{r \times L}.$$

One row is one parity rule; a zero entry means that section does not participate. In systematic form, check c has coefficient one on parity position p_c and sparse coefficients on selected information symbols:

$$p_c = - \sum_{j \in S_c} h_{cj} u_j \pmod{q}.$$

Thus H is small and procedural. The graph support and integer coefficients are currently fixed hyperparameters. They could be searched offline, but ordinary gradients cannot change them while exactly preserving modular validity, systematic encodability, and sparsity. Local bank values and decoder calibration are continuously learnable.

Triadic codes are a special case where a check connects a small number of variables, typically two information symbols and one parity symbol. The generic sparse-linear code allows arbitrary sparse neighbourhoods and coefficients; the triadic case is obtained by setting all other entries of that row to zero.

5 Section D0 from the global D0 algebra

Global D0 used $r = y - \Phi a$ and $g = \Phi^H r$. Substituting the section model gives

$$r^t = y - \sum_{\ell=1}^L F_{\ell} s_{\ell}^t, \quad g_{\ell}^t = \Re\{F_{\ell}^H r^t\}, \quad u_{\ell}^t = s_{\ell}^t + \eta_t g_{\ell}^t.$$

Every device chooses exactly one atom in every section, so

$$\mathbf{1}^T s_{\ell} = K_a.$$

Under uniform section symbols, a coordinate has the exact marginal prior

$$S_{\ell,j} \sim \text{Binomial}(K_a, 1/N_{\ell}).$$

The denoiser evaluates counts $c = 0, \dots, K_a$:

$$\pi(c | u) \propto \binom{K_a}{c} p^c (1-p)^{K_a-c} \exp\left[-\gamma_t(u-c)^2/(2\tau_t) + b_t c\right], \quad p = 1/N_{\ell}.$$

The soft count is $\mathbb{E}[S | u]$ and support evidence is $\log \Pr(S > 0 | u) - \log \Pr(S = 0 | u)$. This one collision-aware route automatically approaches the binary case when multiplicities are negligible.

For any upstream physical gradient g_y ,

$$\frac{\partial \mathcal{L}}{\partial s_{\ell}} = F_{\ell}^H g_y.$$

Backpropagation therefore trains selected local columns and D0 scalars. It cannot train the integer path choices or H without a separate discrete method.

6 Why local recovery is not complete-message recovery

This was the central conceptual failure. With two users and two sections, suppose the receiver perfectly recovers

$$s_1 : \{a, b\}, \quad s_2 : \{c, d\}.$$

The complete messages might be

$$(a, c), (b, d) \quad \text{or} \quad (a, d), (b, c).$$

Both assignments produce identical section counts and hence the identical channel signal $\sum_{\ell} F_{\ell} s_{\ell}$. No local-state decoder can distinguish them unless the allowed path set or some physical cross-section feature breaks the symmetry.

Parity rejects many recombinations, but the multiuser receiver observes a *set of paths*. For example, if both (a, c) and (b, d) are valid and both crossed paths are also valid, the ambiguity remains exactly. More subtly, even when only one pairing is globally valid, single-section marginals do not record which candidate symbols should be associated across sections. High-quality local support is therefore necessary but not sufficient for low complete- message PUPE.

This is different from a global message collision. A global collision means two users chose the same complete B -bit payload and should produce count two. A local collision means two different complete messages chose the same atom in one section. At $B = 100$ the former is negligible at realistic load, while the latter can be common whenever $N_\ell = 2^J$ is modest.

7 Outer BP and valid-path extraction

D0 supplies unary log-evidence for each section symbol. The parity graph has variable nodes $x_\ell \in \mathbb{Z}_q$ and check nodes c . A check-to-variable sum-product update is

$$m_{c \rightarrow \ell}(x_\ell) \propto \sum_{\{x_j: j \in \mathcal{N}(c) \setminus \ell\}} \mathbf{1} \left\{ \sum_{j \in \mathcal{N}(c)} H_{cj} x_j = 0 \pmod{q} \right\} \prod_{j \in \mathcal{N}(c) \setminus \ell} m_{j \rightarrow c}(x_j).$$

Variable updates multiply the unary evidence and all other incoming check messages. The implementation performs these full-alphabet operations differentiably and learns temperature, damping, and check gain.

This is still a single-path marginal inference rule applied to a multiuser superposition. It does not maintain an explicit posterior over sets of K_a associated paths. Consequently, BP can sharpen attractive but mutually incompatible marginals.

Evaluation uses beam search over systematic information symbols. When a parity becomes determined, its belief score is added; each completed information tuple is encoded to its unique valid path. For $B \leq 20$, candidate-path multiplicities are fitted because global collisions matter. At larger B , complete-message collisions are assumed negligible and unique paths are selected.

Hard top- k and beam decisions are not differentiated. Training instead combines local count loss, outer marginal loss, and a contrastive ranking loss that places each true path above random paths and cross-user recombinations. This can improve the rank of a true path so it enters the beam; it does not make the discrete beam search itself smooth. Aggressive top- J pruning early in training can delete the true symbol before useful gradients develop, so no narrow top list is used in the differentiable path.

8 Controlled bridge experiments

8.1 Job 023: small end-to-end pilot

Only 16 of 26 rows returned. Exact-energy triadic banks were stronger than the overlapping sampled-energy row, whose sampled codeword energy reached about 1.29. BP was worse than D0 in eight of nine returned aggregates, and outer-loss gradient norms reached 10^{15} before clipping. The pilot exposed numerical and semantic problems; incomplete rows prevent it from being a final comparison.

8.2 Job 024: large-payload execution without learning success

Every $B = 128$, $J = 16$, $n = 38400$ row completed with $16 \times 65,536$ local states and maximum unit-energy error 4.8×10^{-7} . However, PUPE was 1.0 throughout, support loss stayed near 15, logits saturated at the lower clamp, and initial gradients were 10^{-14} – 10^{-13} .

Evidence status: Exact execution and energy certificate, not a successful communication result. It proves that the tensor shapes scale; it does not prove that the evidence, outer code, or decoder learns at $B = 128$.

8.3 Job 025: the one-component backend is innocent

This job used the same dense, sparse-global, and ODMA physical matrices as the explicit landscape. Global D0 and section-compatible D0 matched exactly at every layer in logits, soft counts, hard counts, and parameter gradients. Separately trained Binomial D0 differed by less than 0.0037 PUPE. The change from a materialised/global operator to the local backend and its collision-aware prior therefore did not cause the later performance collapse.

8.4 Job 026: multiple components expose association

The explicit and local physical signals matched within 6.0×10^{-7} and energy within 6.6×10^{-7} . The high-SNR complete-message PUPE was:

Average over 8 and 12 dB	$K_a = 9$	$K_a = 17$	$K_a = 30$
Identity, local association	.9453	.9692	.9862
Triadic fixed, local association	.5191	.7909	.8945
Triadic learned, local	.4141	.8272	.9469
Triadic learned, BP	.4340	.8465	.9573

Table 1: Job 026. The identity row has no parity factor and is an identifiability control.

Triadic constraints help at $K_a = 9$, proving that outer validity carries useful association information. Performance then collapses as local occupancy rises. With $J = 4$, each section has only 16 atoms, so local collisions are almost certain even when complete-message collisions are modest. Learned banks were optimised mainly through local D0 rather than a set-of-paths objective and generalised poorly at larger load. BP worsened every reported high-SNR triadic cell by 0.007–0.022.

9 What failed, and what was actually learned

The scalable route succeeded in four narrow senses: it removed the explicit M axis, generated legal paths, guaranteed unit energy, and reproduced the explicit physical map on small controls. It failed as a complete current receiver:

1. small local alphabets create heavy local collisions;
2. section counts discard cross-user association;
3. the chosen sparse-linear/triadic outer code restricts the global family before its cost is known;
4. marginal BP does not represent a posterior over an unordered set of multiple paths;
5. local training objectives do not directly reward global set identifiability;
6. large- J D0 calibration saturates before useful gradients emerge.

These results do not prove that every sectioned or procedural code excludes an optimum, or that CCS-AMP itself fails. The implementation borrowed the section/outer-code principle but did not

reproduce a paper-matched AMP algorithm, state evolution, power allocation, or tree decoder. The result is about this encoder/decoder combination.

The experiment also clarified a research-design risk: solving scalability by immediately selecting a particular outer tree can make decoding tractable while removing unknown, possibly better global codebooks. The more general direction is to start from the empirically strong $L = 1$ sparse-global family and generate its supports compactly, only accepting structural restrictions that survive direct controls. Job 027 quantified how sparse that family can become; Report 5 develops the first hash-skeleton candidate.